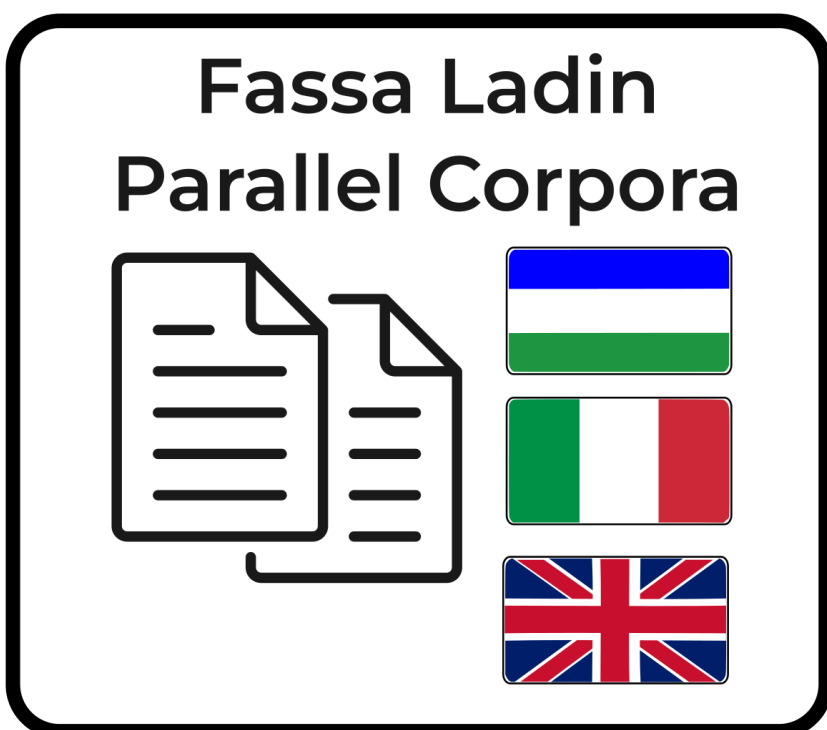
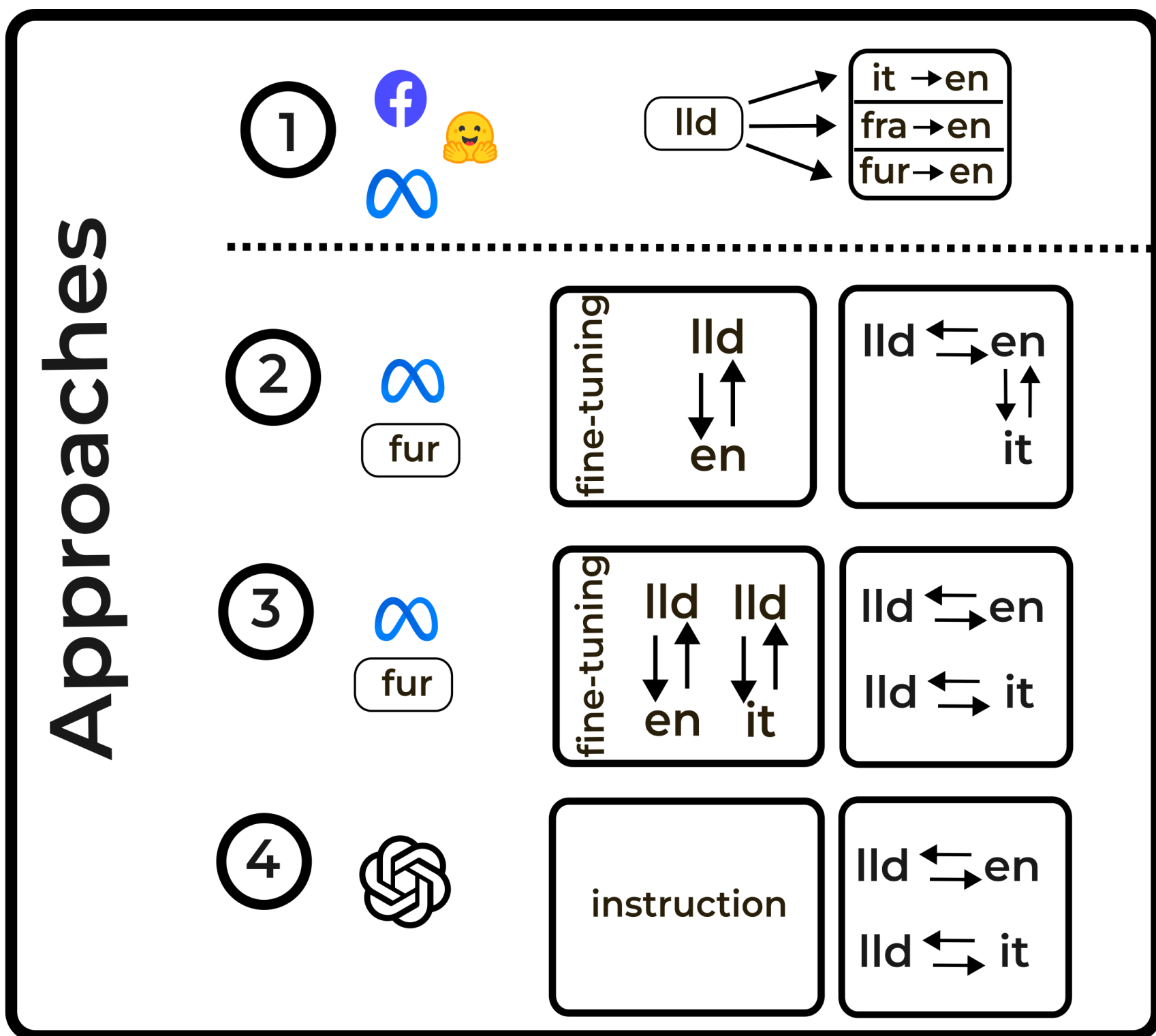


Nesciun Lengaz Lascià Endò: Machine Translation for Fassa Ladin

Giovanni Valer, Nicolò Penzo, Jacopo Staiano



Domains			
Laws	Games	Literature	News
Tourism			
Train		Dev	Test
ID		OOD	



Evaluation

- BLEU
- chrF++
- BERTscore

OPUS-MT, M2M-100, NLLB-200, gpt-4o

Ladin L porta dant azions per didèr dò la medema opotunità anter eles e ic.

Italian Promuove azioni per favorire pari opportunità tra donne e uomini.

English It promotes actions to foster equal opportunities between women and men.

Our dataset

Subset	Original Lang.	Sentences	
Laws	lld, it	742	65,4%
Games	lld, it, en	150	13,2%
Literature	lld, it, en	144	12,7%
News	lld, it	42	3,7%
Tourism	lld, it, en	57	5,0%
Total		1135	100%

Preliminary experiments: it ↔ en

- NLLB-200 outperforms M2M-100 (avg BLEU +8.24)
- Still, unidirectional bilingual models (OPUS-MT) have consistently higher scores wrt. multilingual models

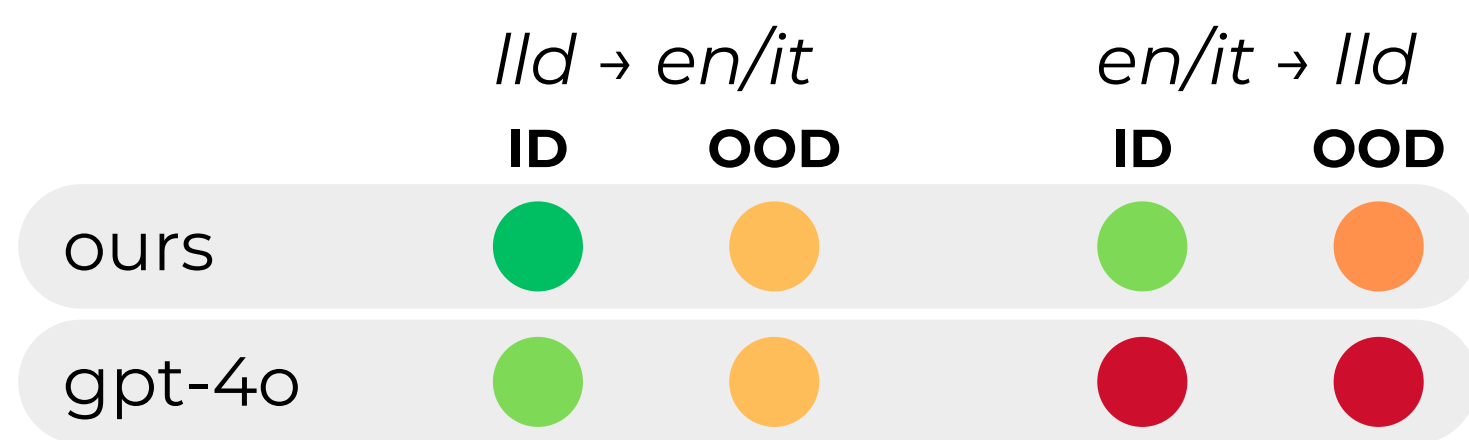
Experiment ①

- NLLB-200 outperforms all other models
- Friulian is the most prone language to transfer learning to Ladin (BLEU):
lld (**ita**) → en: 18.52
lld (**fra**) → en: 13.32
lld (**fur**) → en: **21.76**

Experiments ② ③ ④

- Fine-tuning with the **multilingual translation** approach gives better results than the **zero-shot** pivot-based approach, on en/it → lld (avg BLEU +3.16)
- Gpt-4o has worse performance than our models on the lld → en/it in-domain test (-10.52). However it performs **better** on the **out-of-domain** test (+6.80)
- **Our models can both understand and write in Fassa Ladin.** Gpt-4o cannot: it only understands some Ladin.

English	GT Ladin	MT Ladin	BLEU	chrF++	BERTScore
The national communication campaign on the Electronic Health Record 2.0 also starts in the province of Trento	Ence te la provinzia de Trent pea via la campagna <u>nazionèla</u> de comunicazion <u>sul</u> Fascicol Sanitèr Eletronich 2.0	Ence te la provinzia de Trent pea via la campagna de comunicazion <u>nazionèla</u> <u>su l</u> Fascicol Sanitèr Eletronich 2.0.	62.44	90.22	93.87
The municipal council must adopt rules of procedure governing its functioning.	L Consei de Comun cogn <u>se dèr n</u> regolament <u>che disciplinee</u> so funzionament.	L Consei de Comun cogn <u>aproèr l</u> regolament <u>per</u> so funzionament.	36.96	60.82	86.25
He has an iron stomach.	L'à n <u>stomech</u> de fer <u>acialinà</u> .	<u>El à n</u> <u>stamp</u> de fer <u>acialà</u> .	13.13	42.18	85.50



- Most of our fine-tuning sentences are drawn from laws. This results in **very high performance in the law domain**, but **struggles in other domains**.
- Low scores are not always caused by bad translations, rather by acceptable alternatives.

FINDINGS

- By leveraging the **similarities with a close language** (as Friulian for Ladin), it is possible to fine-tune a pre-trained MT model with **very little data from a specific language variety** (Fassa Ladin).
- Fine-tuning with **more than two languages** (parallel sentences in Fassa Ladin, Italian and English) regularizes the training, providing **better translations**.



Reference:

Giovanni Valer, Nicolò Penzo, Jacopo Staiano. 2024. Nesciun Lengaz Lascià Endò: Machine Translation for Fassa Ladin. In Proceedings of the Tenth Italian Conference on Computational Linguistics, Pisa, Italy.

PAPER



REPO



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